Hasso Plattner Institut
Digital Engineering · Universität Potsdam

Introduction to Probabilistic Machine Learning

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Real-World Applications

Overview

1. adPredictor: Bayesian Probit in e-Commerce
2. MatchBox: Bayesian Recommendation Systems
3. The Path of Go: Bayesian Pattern Ranking for Games

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Predicting Clickthrough-Rate (CTR)

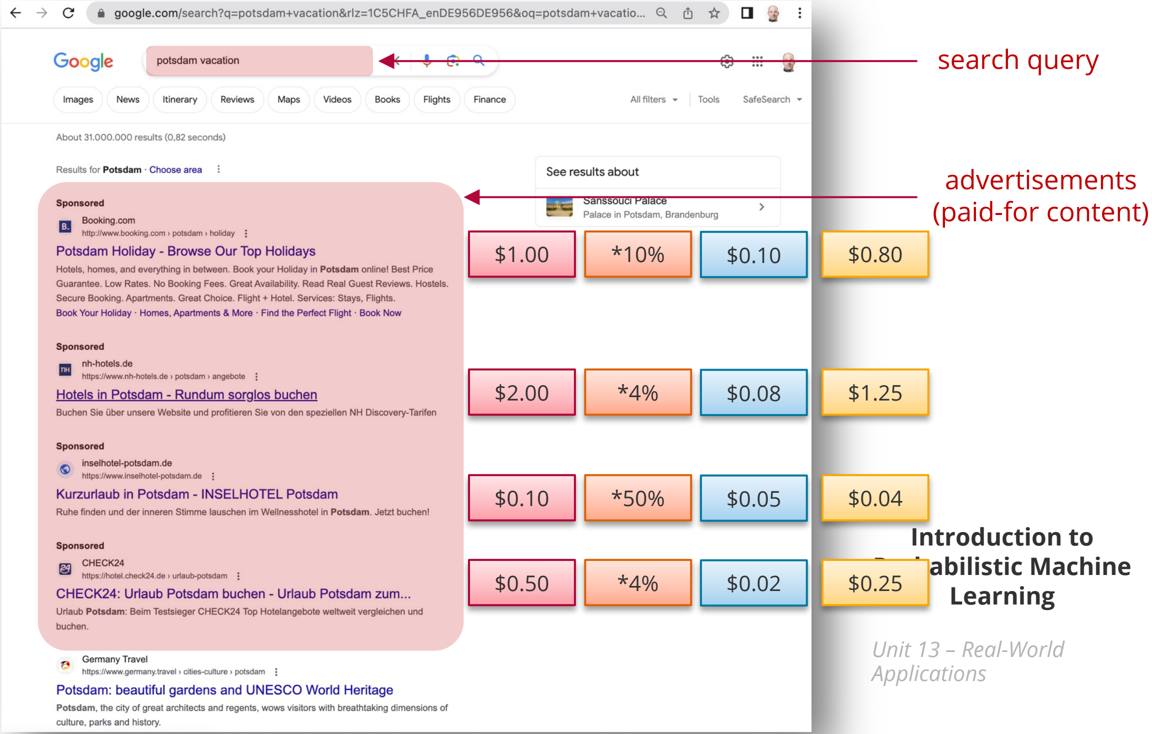
- **Display logic:**

  $$b_1 \cdot p_1 \geq b_2 \cdot p_2 \geq \cdots$$

- **Charge logic:**

  $$c_i = b_{i+1} \cdot \frac{p_{i+1}}{p_i}$$

- **Advantages of improved probability estimates:**
  - Increase user satisfaction by better targeting
  - Fairer charges to advertisers
  - Increase revenue by showing ads with high click-thru rate



Bayesian Probit Regression with Factor Graphs

- **Learning algorithm speed requirement**:
  - 5,000-10,000 impression updates / sec = 100-200 µs per impression update
  - **Decision**: Use one-hot encoding for all raw input data values and features
- **Bayesian Probit model**:

$$\underbrace{p(\mathbf{w}|Y=+1)}_{\text{posterior}} \propto \underbrace{\left[\int_0^\infty \mathcal{N}\left(t; \sum_{k=1}^{D} w_k, \beta^2\right) \mathrm{d}t\right]}_{\text{likelihood}} \cdot \underbrace{\left[\prod_{k=1}^{D} \mathcal{N}\left(w_k; \mu_k, \sigma_k^2\right)\right]}_{\text{prior}}$$

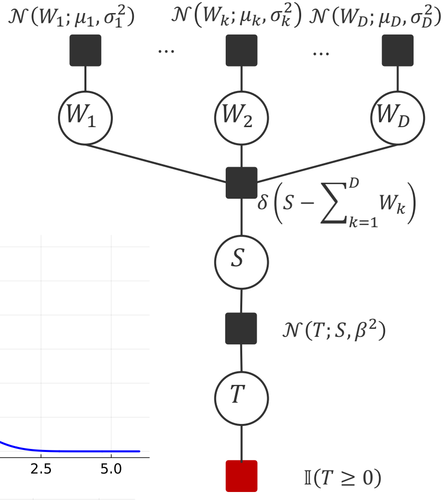
- Closed form using **message passing**:

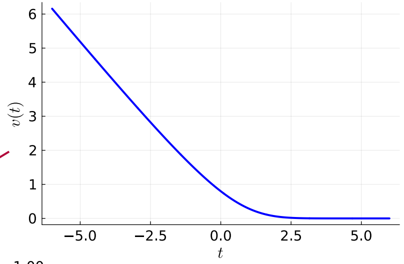
$$\mu_k \leftarrow \mu_k + \frac{\sigma_k^2}{s} \cdot v\left(\frac{\sum_{j=1}^{D} \mu_j}{s}\right)$$

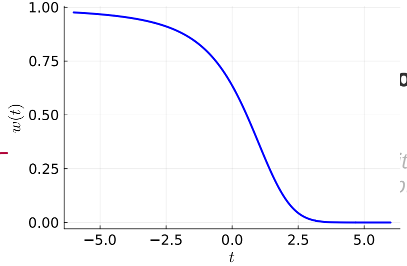
largest for parameter with largest uncertainty so far

$$\sigma_k^2 \leftarrow \sigma_k^2 \cdot \left(1 - \frac{\sigma_k^2}{s^2} \cdot w\left(\frac{\sum_{j=1}^{D} \mu_j}{s}\right)\right)$$

$$s^2 = \beta^2 + \sum_{j=1}^{D} \sigma_j^2$$







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it 13 – Real-World
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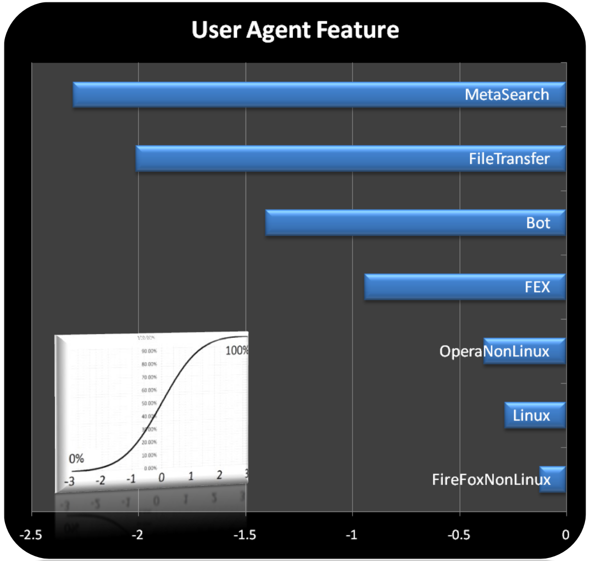
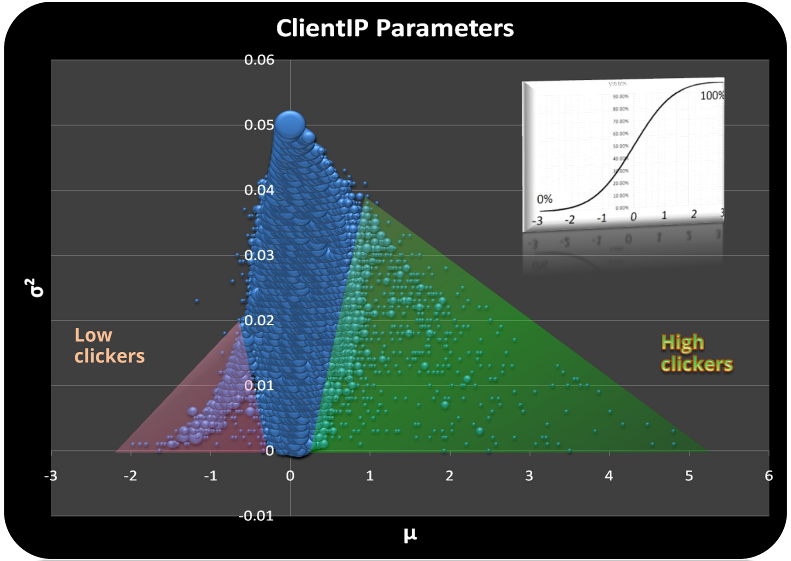
Empirical Findings (bing.com)





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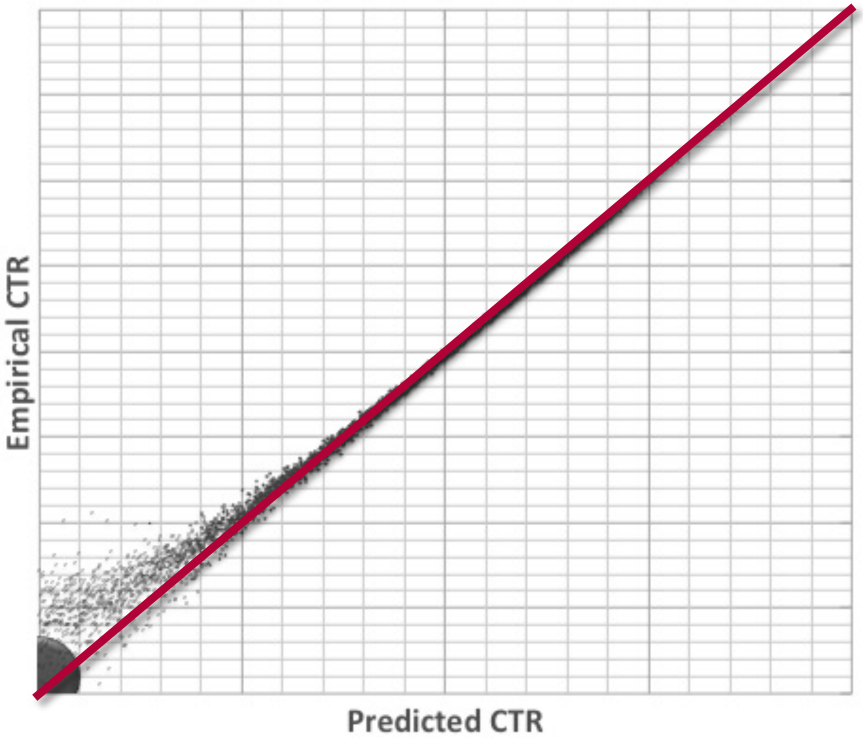
Unit 13 – Real-World Applications

Accuracy



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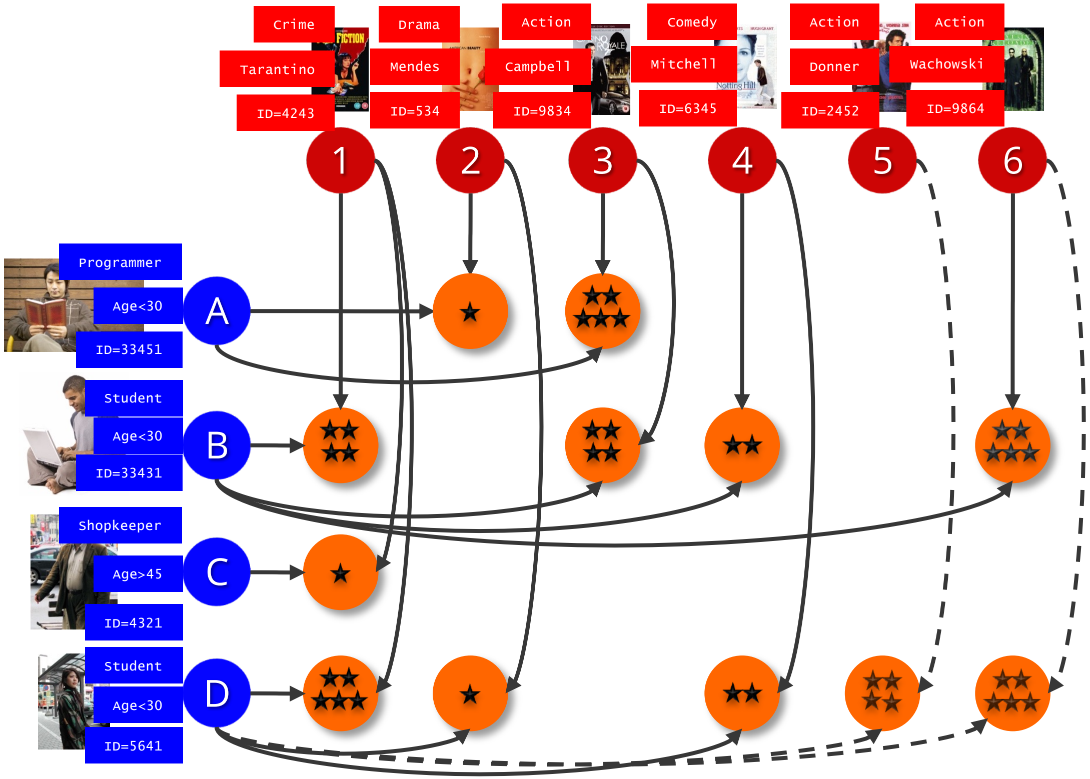
MatchBox: The Recommendation Setting



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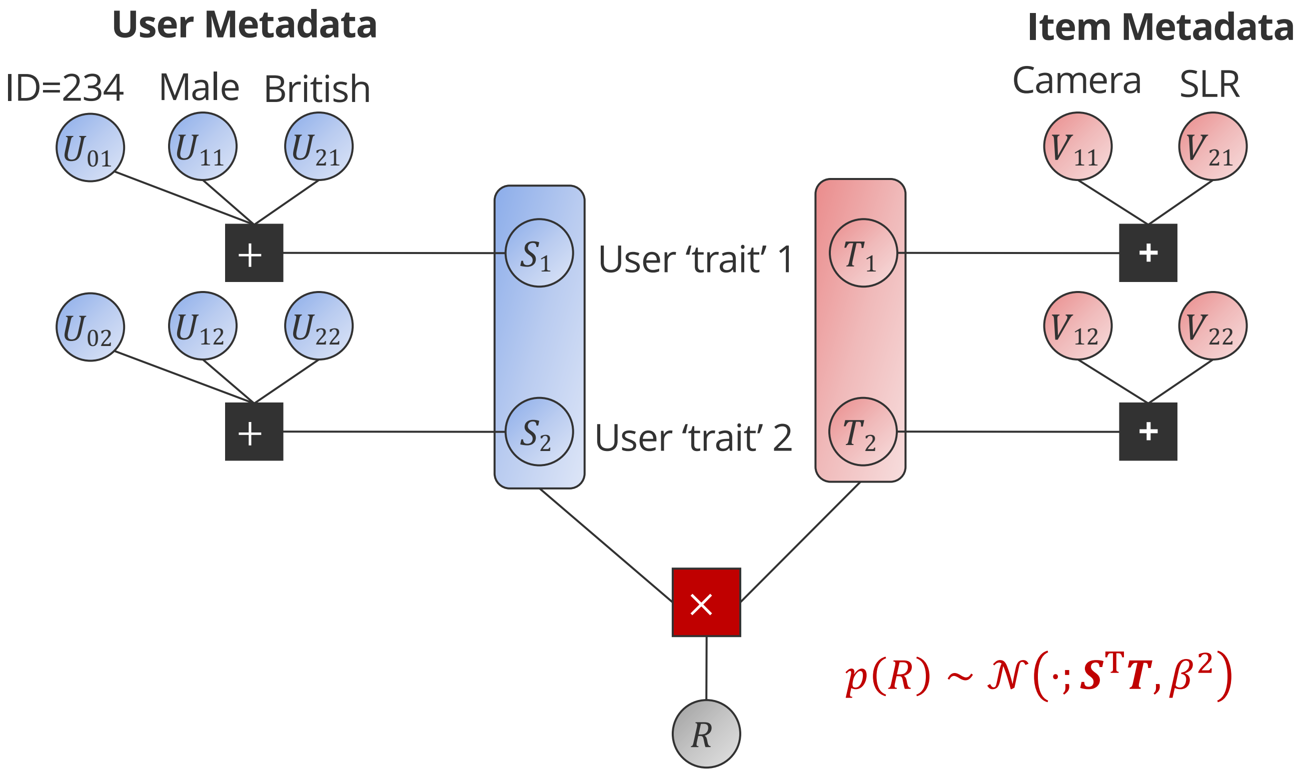
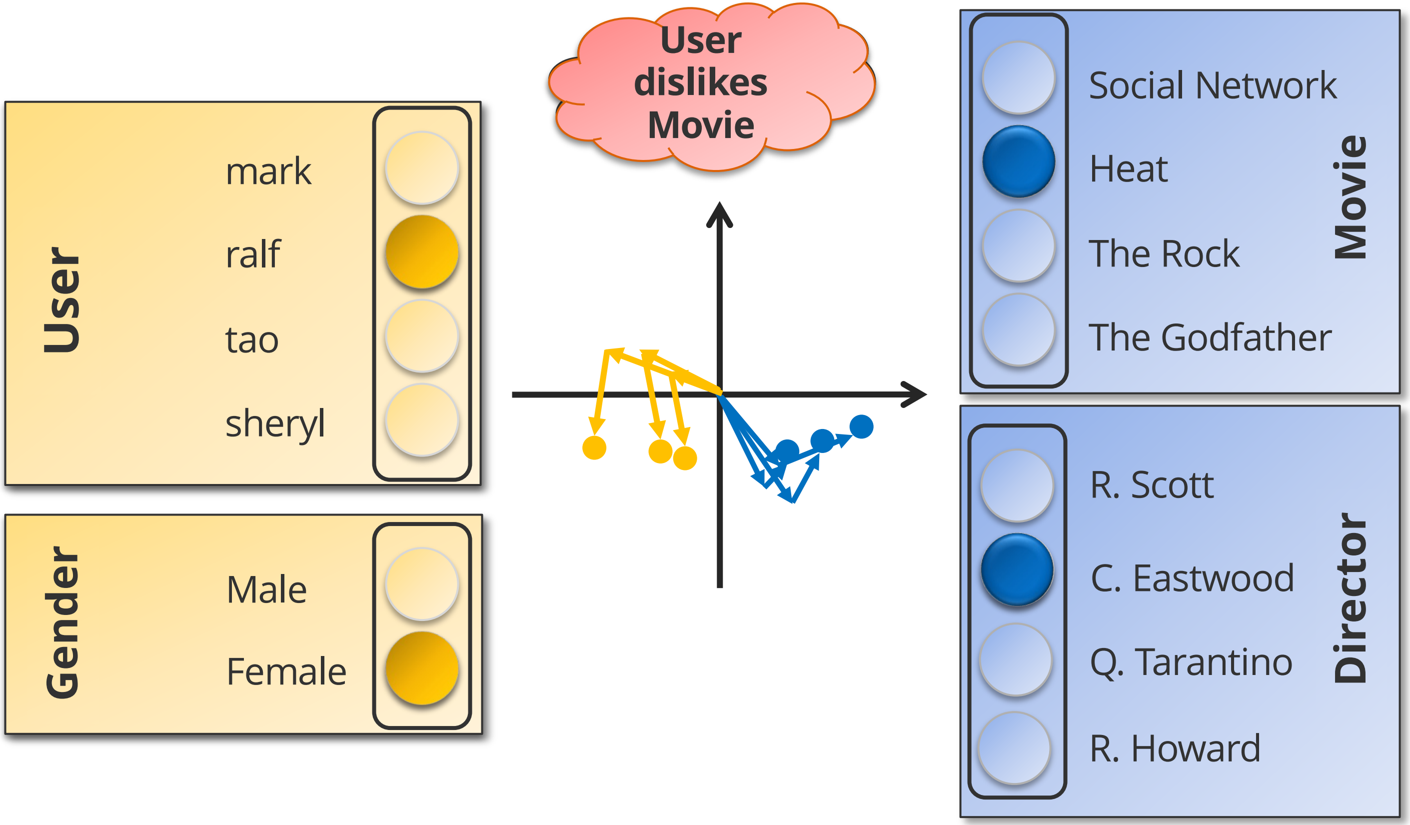
Unit 13 – Real-World Applications

MatchBox With Metadata

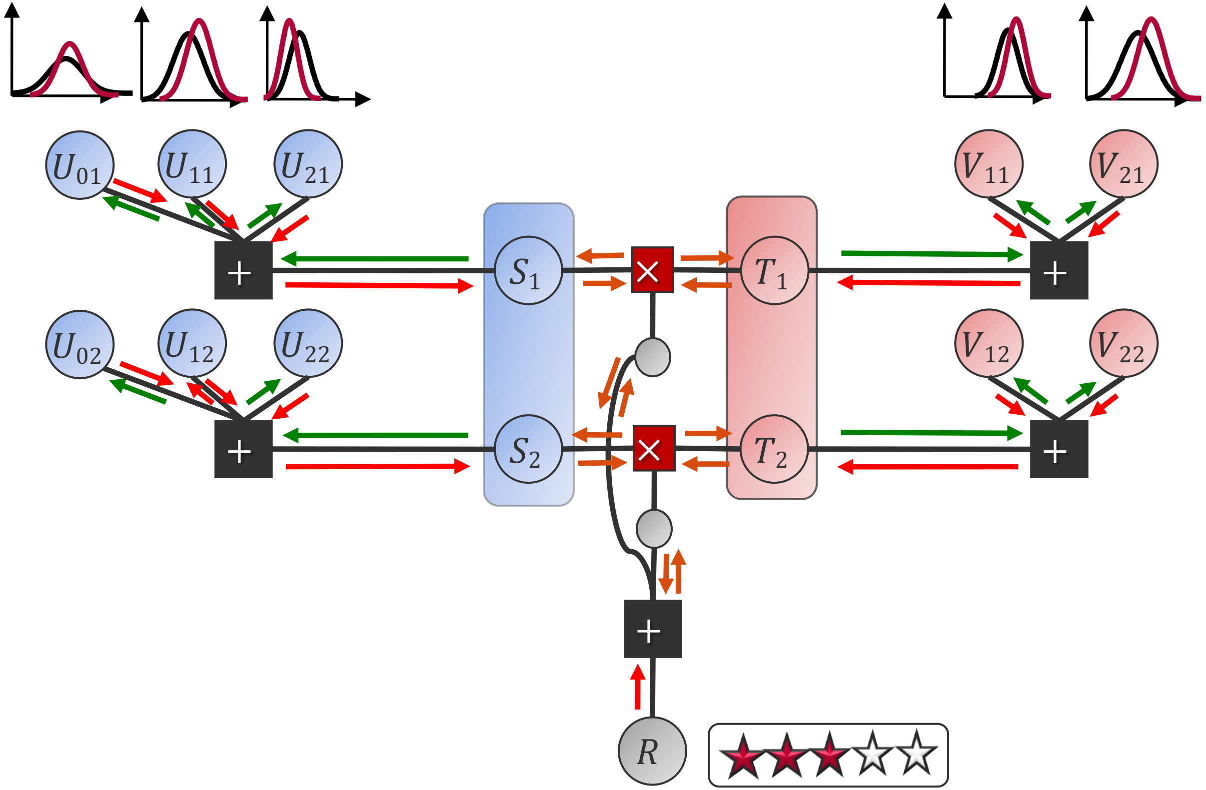


$$p(R) \sim \mathcal{N}\left(\cdot; \boldsymbol{S}^{\mathrm{T}}\boldsymbol{T}, \beta^2\right)$$

Recommender System: MatchBox



Message Passing For Matchbox



MovieLens (1,000,000 Ratings)

6,040 users

User ID

User Job	
Other	Lawyer
Academic	Programmer
Artist	Retired
Admin	Sales
Student	Scientist
Customer Service	Self-Employed
Health Care	Technician
Managerial	Craftsman
Farmer	Unemployed
Homemaker	Writer

User Age
<18
18-25
25-34
35-44
45-49
50-55
>55

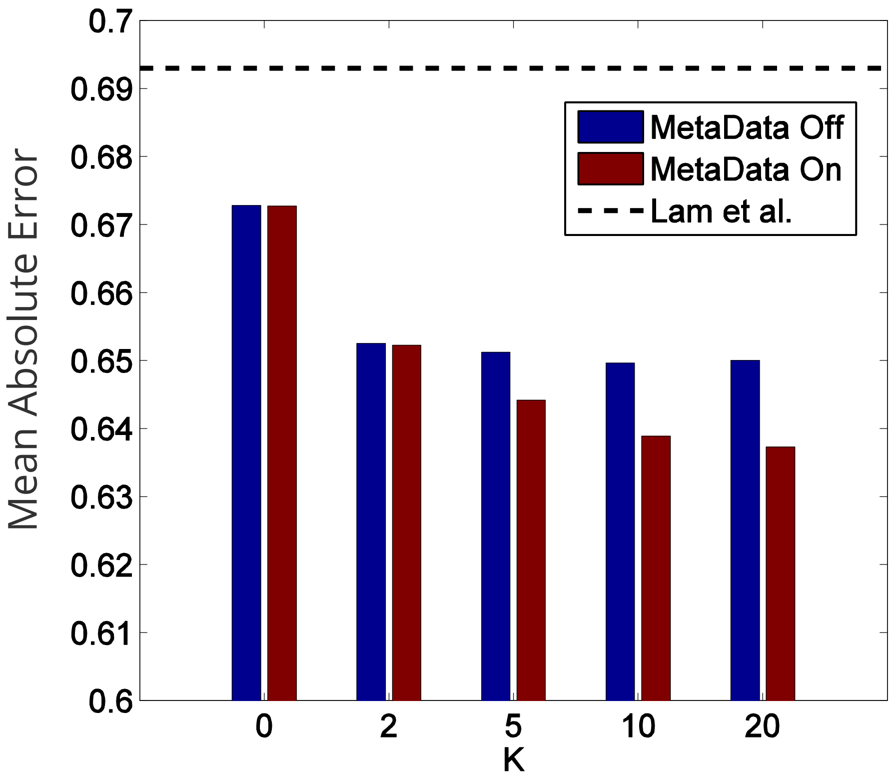
User Gender
Male
Female

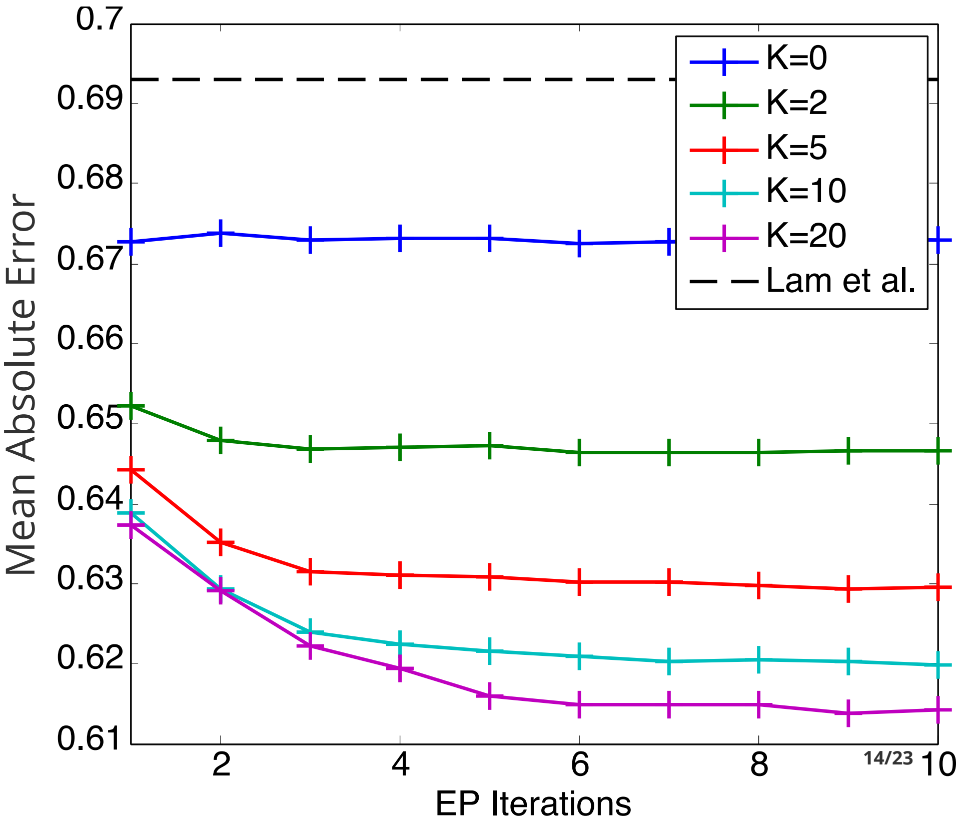
3,900 movies

Movie ID

Movie Genre	
Action	Horror
Adventure	Musical
Animation	Mystery
Children's	Romance
Comedy	Thriller
Crime	Sci-Fi
Documentary	War
Drama	Western
Fantasy	Film Noir

MovieLens with Thresholds Model





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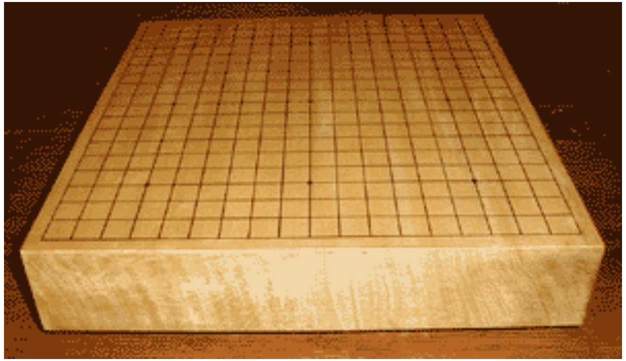
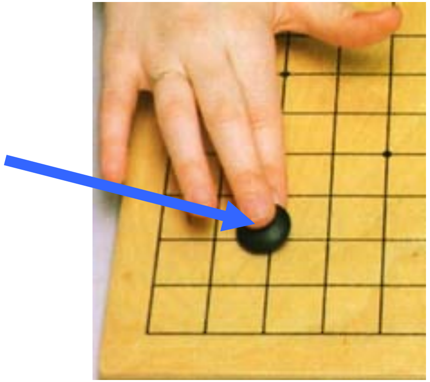
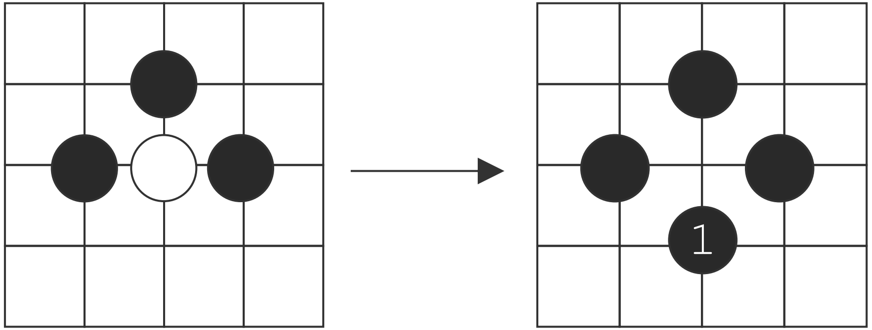
The Game of Go

- Started about 4000 years ago in ancient China.
- About 20 million players worldwide.
- **2 Players**: Black and White.
- **Board**: 19×19 grid.
- **Rules**:
  - Turn: One stone placed on vertex.
  - Capture.
- **Aim**: Gather territory by surrounding it.

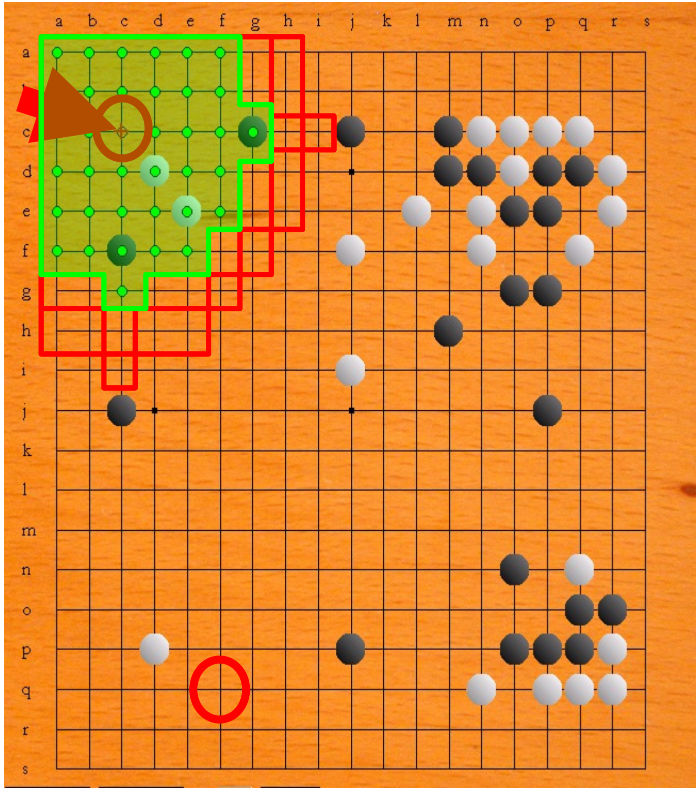






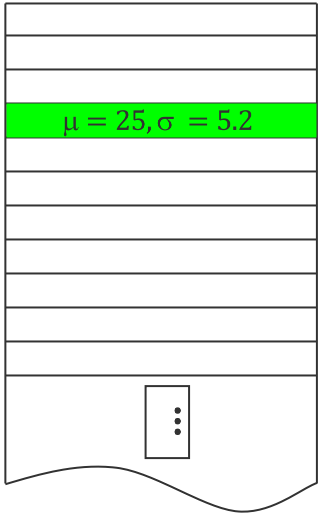
A stone is **captured** by completely **surrounding** it.

Moves Selection by Pattern Matching



Pattern Urgency Table

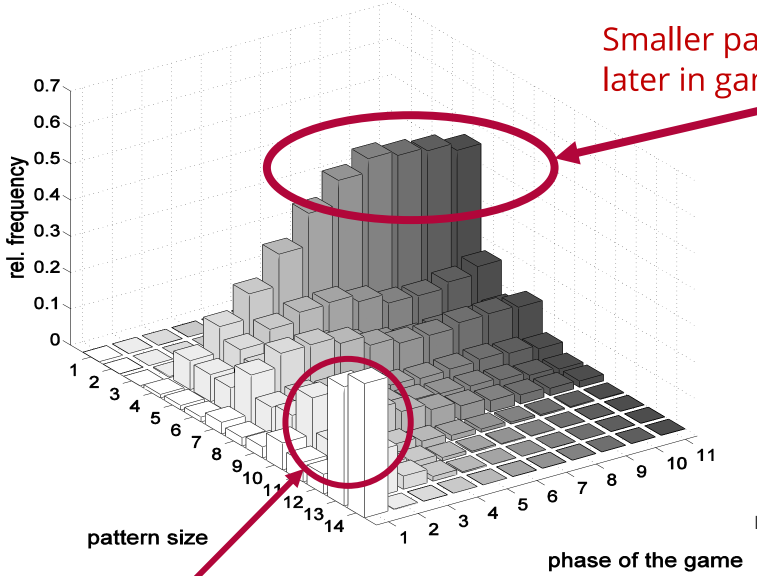
Not in database!



Black to move

Harvesting and Learning

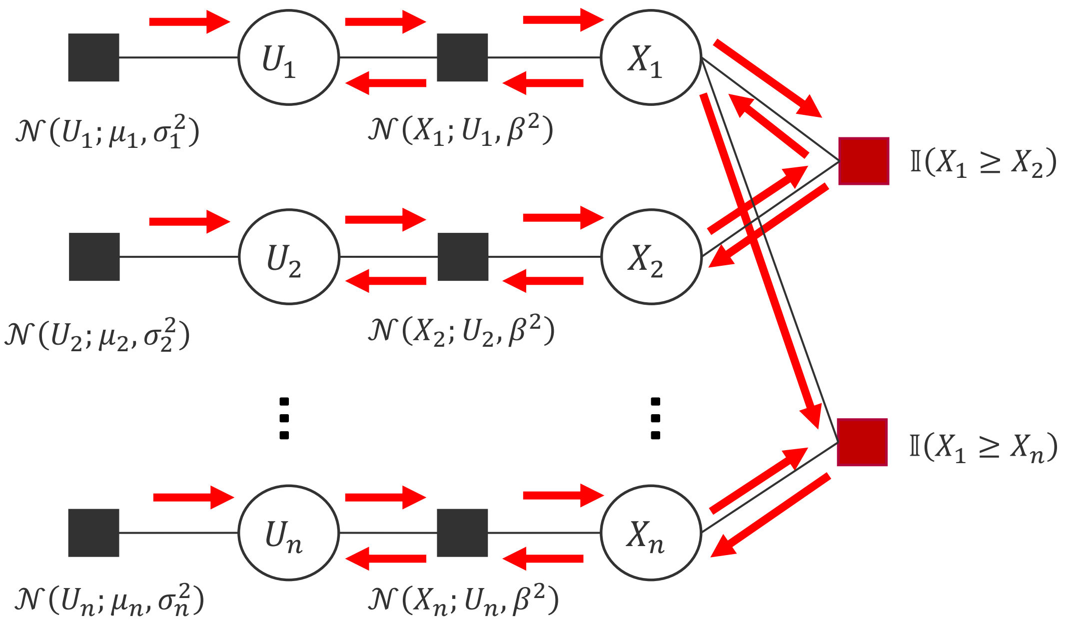
- **Two processes**:
  1. Harvesting patterns (180,000 experts games → 600M possible patterns!)
  2. Ranking patterns (requires model of moved selection)
- **Move selection model** as partial ranking using *urgency*
  - **Move made**: “Wins” over any other move available on the board indicating that it is most *urgent*.
  - **Moves not** made. Nothing can be concluded about the ranking among the available but un-played moves.



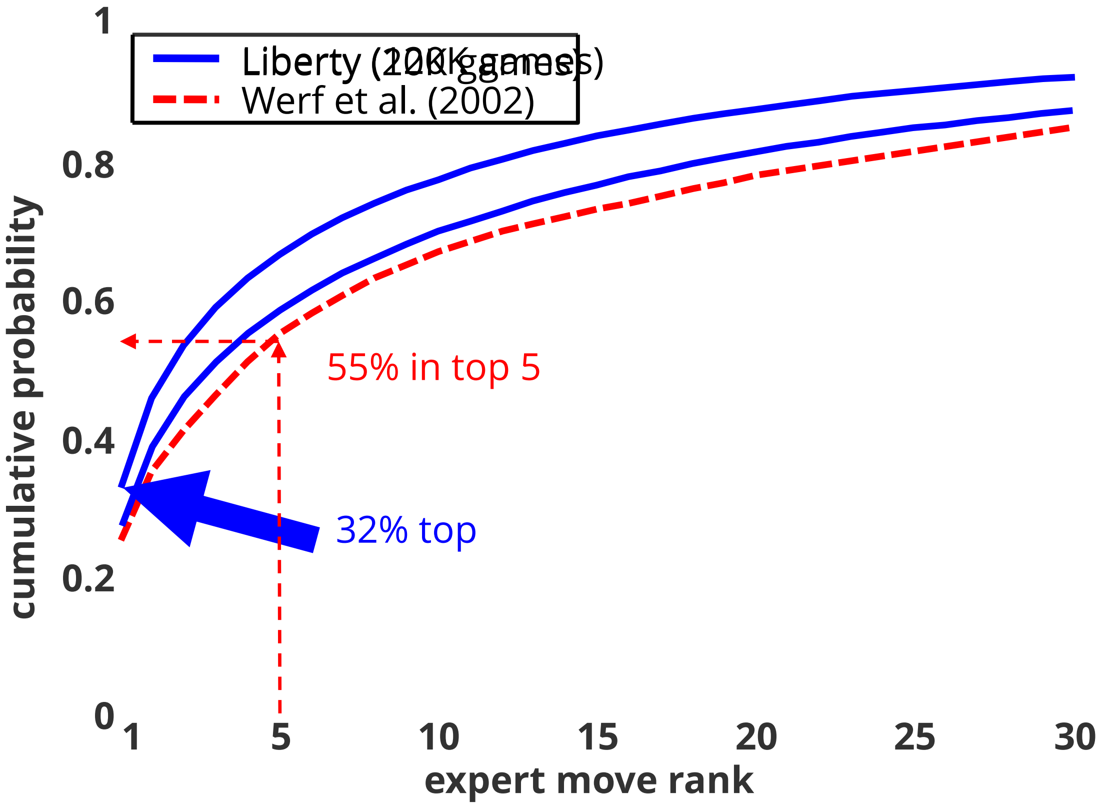
Smaller patterns matched later in game.

Big patterns matched at beginning of game

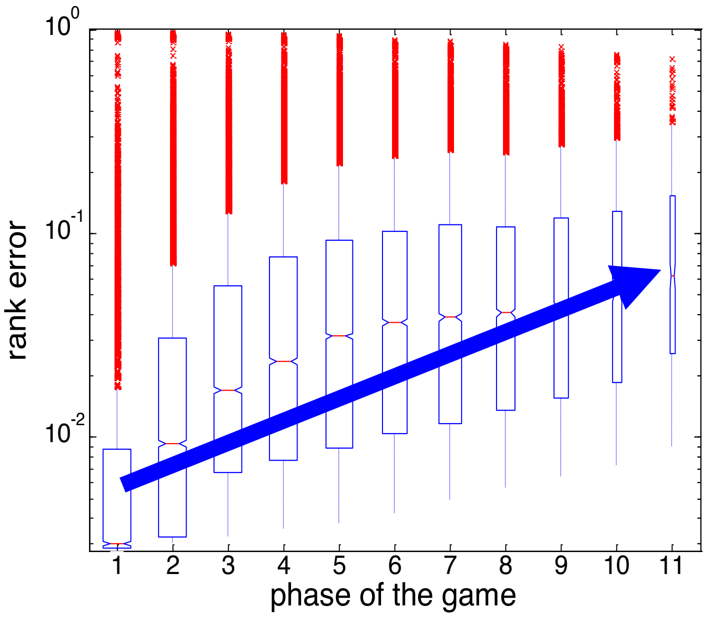
Partial Ranking Message Passing



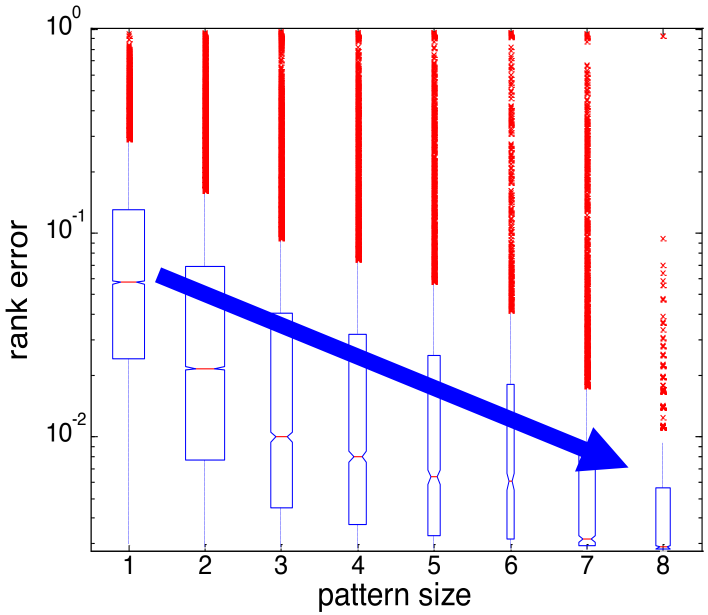
Experimental Results



Experimental Results (ctd.)



Error increases towards end of the game



Error decreases for increasing pattern size

Summary

- **Real-World Applications**
  - Bayesian Probit regression using expectation propagation is used in over 200 production systems at Amazon today (and many more places around the world)
  - Bayesian Probit regression is like a two-team game where one team is the "threshold-0 player"
  - When using the Gaussian product factor, we can even model bi-linear models used in recommendation → requires a different approximation due to multi-modality
  - Partial rankings are another application of the pairwise difference factor
  - Used today in AlphaGo!

Thank You!